

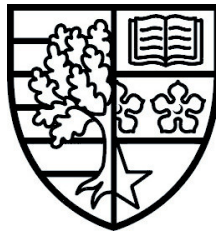
Interactive Gantt Chart Builder

by Fraser DAVIES

Research Report

Submitted in partial fulfilment for the degree of
BSc (Hons) Computer Science

Supervised by Prof. Pierre LE BRAS



HERIOT-WATT UNIVERSITY
School of Mathematical and Computer Sciences

13th November 2025

The copyright in this dissertation is owned by the author. Any quotation from the dissertation or use of any of the information contained in it must acknowledge it as the source of the quotation or information.

AUTHORSHIP DECLARATION

Course code and name	F20PA - Honours Project
Type of assessment	Individual
Coursework title	Interactive Gantt Chart Builder
Student name	Fraser DAVIES
Student ID	H00403114

By signing this form:

- I declare that the work I have submitted for individual assessment or the work I have contributed to a group assessment is entirely my own. I have not taken the ideas, writings or inventions of another person and used them as if they were my own. My submission or my contribution to a group submission is expressed in my own words. Any uses made within this work of the ideas, writings or inventions of others, or of any existing sources of information (books, journals, websites, etc.) are properly acknowledged and listed in the references and/or acknowledgements sections.
- I confirm that I have read, understood and followed the University's Regulations on academic misconduct as published on the [University's website](#), and that I am aware of the penalties that I will face should I not adhere to the University Regulations.
- I confirm that I have read, understood and avoided the different types of plagiarism explained in the University guidance on [Academic Integrity and Plagiarism](#).

Student signature: Fraser DAVIES

Date: 13th November 2025

Your work will not be marked if a signed copy of this form is not included with your submission.

ACKNOWLEDGEMENTS

Acknowledgements text.

ABSTRACT

Abstract text.

TABLE OF CONTENTS

Authorship Declaration	i
Acknowledgements	ii
Abstract	iii
Table of Contents	iv
1 Introduction	1
1.1 Background and Motivation	1
1.2 Aim and Objectives	2
1.3 Organisation	2
2 Background Research	3
2.1 Gantt Charts and Project Management.	3
2.2 Gantt Charts in Educational Contexts	4
2.3 Interactive Data Visualisation	4
3 Methodology	4
3.1 Requirement Analysis	4
4 Feasibility	4
References	5
A Project Management	6
A.1 Gantt Chart	6
B Generative AI	6

1 Introduction

1.1 Background and Motivation

Project management is critical for students in higher education, particularly those undertaking substantial individual projects such as honours dissertations. Dissertations present unique challenges as they are often the largest and most complex projects students have encountered, requiring sustained effort over extended periods while juggling multiple responsibilities [8]. Effective planning, scheduling, and progress monitoring are essential skills that students must develop to successfully navigate these complex, long-term academic endeavours. However, many students struggle with translating abstract project requirements into concrete, manageable tasks with realistic timelines and dependencies, finding themselves overwhelmed by the scope and duration of dissertation work [8].

Gantt charts have emerged as one of the most widely adopted project management tools, originally developed by Henry Gantt in the early 20th century to visualise project schedules through horizontal bar representations [5, 10]. These visual tools effectively display task durations, dependencies, and timelines, making them invaluable for planning and tracking project progress [3]. In educational contexts, Gantt charts serve as pedagogical tools that help students manage their increasing responsibilities while developing time management skills. Their graphical representation visualises beginning dates, ending dates, and duration parameters, helping to track various projects over specific time periods [2].

Despite their proven utility in professional project management, traditional Gantt chart tools often present usability challenges for students who lack prior project management experience. Many existing solutions are designed for team-based projects and enterprise environments, making them overly complex for individual academic projects. Furthermore, the widespread adoption of visualisation tools in educational settings has been hindered by usability problems and the time required for instructors and students to learn and incorporate these tools into their workflows.

At Heriot-Watt University (HWU), honours project students are currently required to submit project management plans as part of their dissertation deliverables. However, the existing project management system lacks integrated tools that guide students through the process of creating well-structured Gantt charts with appropriate task hierarchies, dependencies, and milestones. This gap in the current system motivated the development of an interactive Gantt chart builder specifically tailored to the needs of honours project students and their supervisors.

1.2 Aim and Objectives

The primary aim of this project is to design, develop, and evaluate an interactive Gantt chart builder that will be integrated into the existing HWU honours project management system. This tool will provide structured guidance to help students create comprehensive project plans while offering supervisors visibility into student progress and planning quality.

The specific objectives of this project are:

- (1) To conduct comprehensive background research on Gantt chart design principles, project management methodologies, and interactive visualisation techniques relevant to educational contexts
- (2) To gather and analyse requirements from key stakeholders, including current system developers, supervisors, and students, to ensure the tool meets the needs of its target users
- (3) To design an intuitive user interface that guides students through the process of defining milestones, tasks, task hierarchies, and dependencies without requiring prior project management expertise
- (4) To develop an interactive Gantt chart visualisation using D3.js (Data-Driven Documents), a powerful JavaScript library that enables the creation of dynamic and interactive data visualisations in web browsers using HTML, SVG, and CSS [4, 11], ensuring seamless integration with the existing HWU project system's technological stack
- (5) To implement functionality that allows students and supervisors to explore project timelines, track progress, and reevaluate goals as projects evolve
- (6) To provide export capabilities that enable students to include their Gantt charts and associated data in their project documentation
- (7) To conduct a rigorous usability evaluation through user studies with students and supervisors to assess the effectiveness of the tool and its guidance features

1.3 Organisation

The organisation of this research report is as follows:

- Chapter 1: Introduces the project, outlining the background and motivation, aim and objectives, and the organisation of the report.
- Chapter 2: Presents the background research, discussing related work found in the technical literature and its relevance to the project.

2 Background Research

2.1 Gantt Charts and Project Management

Building upon Henry Gantt's original design, modern Gantt charts have evolved to incorporate sophisticated features that support complex project planning. A comprehensive Gantt chart integrates multiple elements: [3, 10]

- A vertical list of tasks or activities on the left side.
- A horizontal timeline indicating project duration (days, weeks, or months)
- Graphical bars representing individual task durations and their temporal placement.
- Lines or arrows indicating dependencies between tasks.
- Labels showing team members or assignees.
- Special markers (typically diamonds) that denote milestones or important project achievements.

Task dependencies are fundamental to effective Gantt chart usage, defining the relationships between tasks and dictating the order in which activities must occur. Project management identifies four primary dependency types [7, 9]. The Finish-to-Start (FS) dependency, where a successor task cannot commence until its predecessor concludes, represents the most common relationship type [6]. Start-to-Start (SS) dependencies allow a task to begin only after another task has also begun, enabling parallel workflows that share a start point. Finish-to-Finish (FF) relationships constrain task completion, requiring one task to finish before another can conclude. Finally, the Start-to-Finish (SF) dependency, the least common type, denotes that a task cannot finish until another task begins, typically appearing in handover or transition scenarios.

Beyond dependencies, Gantt charts visualise the critical path the longest sequence of dependent tasks that determines the project's minimum completion time. This enables project managers to distinguish between tasks that are critical to meeting deadlines and those that possess scheduling flexibility [1]. This distinction proves particularly valuable for resource allocation and risk management decisions.

2.2 Gantt Charts in Educational Contexts

2.3 Interactive Data Visualisation

3 Methodology

3.1 Requirement Analysis

4 Feasibility

References

- [1] Asana. 2025. What is a Gantt Chart? Guide to Project Timelines. <https://asana.com/resources/gantt-chart-basics>. Accessed: 12/11/2025.
- [2] Andrea Bednjanec and Martina Filipovic Tretinjak. 2013. Application of Gantt charts in the educational process. In *2013 36th International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)*. MIPRO, 631–635.
- [3] Academy for International Modern Studies (AIMS). [n. d.]. What is Gantt Chart Project Management? <https://aims.education/what-is-gantt-chart-project-management/>. Accessed: 08/11/2025.
- [4] GeeksforGeeks. 2025. *D3.js in JavaScript*. <https://www.geeksforgeeks.org/javascript/d3-js/> Accessed: 25/09/2025.
- [5] GeeksforGeeks. 2025. What is Gantt Chart in Project Management? <https://www.geeksforgeeks.org/business-studies/what-is-gantt-chart-in-project-management/>. Last updated: 23 Jul 2025. Accessed: 08/11/2025.
- [6] Adrian Neumeyer. [n. d.]. Finish-to-Start Dependency Explained (with Examples). <https://www.tacticalprojectmanager.com/finish-to-start-dependency/>. Accessed: 12/11/2025.
- [7] ProjectManager. 2021. Gantt Chart Dependencies: Understanding Task Dependency Types. <https://www.projectmanager.com/blog/gantt-chart-dependencies>. Accessed: 12/11/2025.
- [8] Katie Shives. 2015. *Using Project Management Approaches to Tame Your Dissertation*. <https://www.insidehighered.com/blogs/gradhacker/using-project-management-approaches-tame-your-dissertation> Accessed: 08/11/2025.
- [9] Smartsheet. 2020. *Gantt Chart with Dependencies*. <https://www.youtube.com/watch?v=U4QftGiN5Nk> Uploaded on YouTube by Smartsheet. Accessed: 30/09/2025.
- [10] Coursera Staff. 2025. Gantt Charts: What They Are and How to Make Them. <https://www.coursera.org/articles/gantt-charts>. Accessed: 08/11/2025.
- [11] W3Schools. 2025. *D3.js Tutorial*. https://www.w3schools.com/js/js_graphics_d3js.asp Accessed: 25/09/2025.

A Project Management

A.1 Gantt Chart

B Generative AI